
DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

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ABSTRACT

AI weather models are often reported to underestimate extremes, but most evidence concerns deterministic regression models verified against reanalysis. We evaluate twelve physical and AI forecast models against ECMWF IFS using ten months of European station observations. The evaluation covers 10 m wind, 2 m temperature, solar radiation, and precipitation within regimes defined from a fixed ERA5 1991–2020 climatology. We find no uniform AI-specific deficit in the tails. Several AI models remain more accurate than IFS under extreme conditions, while others deteriorate markedly; comparable variation occurs among physical models. Every model nevertheless exhibits a common conditional-error pattern, overpredicting low observations and underpredicting high observations. Attenuation of extreme values therefore does not imply a uniform loss of relative skill: tail performance depends on the model, variable, and evaluation setting rather than on whether the forecast is produced by AI or physical numerical modelling.

1 INTRODUCTION

Data-driven weather models have achieved strong performance on standard forecast metrics, often matching or surpassing operational numerical weather prediction (NWP) models (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). However, many of these models are trained using mean-squared error and tend to regress towards the conditional mean, smoothing small-scale variability and underestimating extremes, as reported for peak winds and record-breaking events (Olivetti & Messori, 2024a; Bonavita, 2024; Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

The available evidence on this behaviour remains nevertheless narrow. Most existing literature concerns deterministic regression models evaluated against reanalysis. Probabilistically trained models, generative ensembles, and high-resolution regional models have received less attention, especially when verified against weather stations, where errors can be substantially larger (Jin et al., 2024; Gupta et al., 2025; Jua AI, 2025c). Among these, generative models are particularly relevant because they represent a distribution of possible outcomes rather than a single conditional estimate, and recent work has shown promising performance for weather extremes and high-resolution regional forecasting (Price et al., 2025; Pathak et al., 2026; Molinaro et al., 2024).

A further limitation is that underestimation is often attributed specifically to AI, while NWP models may also underestimate the same observations. Hence, the relevant question is not whether AI models underestimate extremes, but whether they are less accurate than NWP when extremes occur. An AI model can underestimate an extreme observation while remaining closer to it than an NWP forecast.

We test this using twelve forecast models, and compare against ECMWF IFS as the reference. Forecasts are evaluated over ten months against European synoptic, solar-radiation, and rain-gauge observations. The analysis covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly precipitation at lead times up to 48 h. We evaluate extremes in two complementary ways: first, by stratifying forecasts according to locally defined climatological regimes; and second, through externally documented record-breaking temperature events and major European windstorms.

Our main contributions are:

- We extend station-based evaluation of weather extremes to a wide range of NWP and AI weather models, including deterministic regression models, CRPS-trained and perturbations-based global ensembles, and high-resolution regional generative models.
- We distinguish conditional forecast errors from relative accuracy, testing whether AI models that underpredict extreme observations are also less accurate than the NWP reference under the same conditions.
- We evaluate performance across four surface variables, forecast leads, countries, and model classes, and complement the climatological analysis with externally documented temperature-record events, major windstorms, and heavy-precipitation events.

METHODS

We begin by describing the forecast models and station observations, then we introduce the climatological regimes, bias correction, and evaluation protocol. Supporting definitions, robustness checks, and complete numerical results are organised in Appendix A.

2.1 MODELS

Let $u(t)$ denote the atmospheric state at time t . Weather forecasting amounts to approximating its evolution from an initial state $u(t_0)$,

$$\hat{u}_m(t_0 + \tau) = \mathcal{F}_m^\tau(u(t_0)), \quad (1)$$

where m indexes the forecast model and τ is the forecast lead time. NWP models approximate $\mathcal{F}_m^\tau$ through numerical integration of the governing equations on a discrete grid, whereas AI models learn this mapping from historical atmospheric states. Forecasts may be deterministic, consisting of a single state, or probabilistic, consisting of an ensemble,

$$\left\{ \hat{u}_m^{(k)}(t_0 + \tau) \right\}_{k=1}^K.$$

We evaluate twelve forecast models against the high-resolution deterministic ECMWF IFS forecast, which serves as the reference (Table S1; Appendix A.1). The comparison spans NWP and AI models, global and regional domains, and deterministic and ensemble forecasts.

The NWP models are ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models include ECMWF AIFS (Lang et al., 2024a), Microsoft Aurora (Bodnar et al., 2025), and Jua EPT-2 Reasoning (Jua AI, 2025b), all global deterministic regression models. The global ensembles are ECMWF AIFS ENS (Lang et al., 2024b), a 51-member ensemble trained with a continuous ranked probability score objective and learned noise injection, and Jua EPT-2e (Jua AI, 2025b), a perturbation-based ensemble. The regional models are Jua EPT-2 HRRR (Jua AI, 2025a) and EPT-2.1 Europa (Jua AI, 2026a), generative ensembles based on the methodology of Molinaro et al. (2026; 2024), and EPT-2.1 Helios (Jua AI, 2026b), a solar-specialised regional model.

Models are evaluated only for variables with compatible outputs. Precipitation is available from the NWP models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS, AIFS ENS, and Aurora are restricted to wind and temperature because their radiation and precipitation outputs cannot be matched to the hourly verification data.

2.2 STATION OBSERVATIONS

Forecasts are verified against three complementary in situ datasets: roughly operational surface observations for 10 m wind and 2 m temperature, 500 solar-radiation observations for hourly global horizontal irradiance, and observations from 4,005 rain gauges for hourly precipitation. Specific contributing networks and data providers are listed in Appendix A.1.

2.3 OBSERVED-VALUE REGIMES

Regimes are defined independently of the evaluation period using a fixed ERA5 1991–2020 climatology (Hersbach et al., 2020). ERA5 fields are bilinearly interpolated from the 0.25° grid to each station. For each station, calendar day, and UTC hour, the reference sample comprises values from the same UTC hour within a cyclic ± 15 -day window over the 30-year period, yielding 930 values before variable-specific filtering.

Percentile thresholds are computed separately from each reference sample. Wind and temperature use all reference values, whereas solar radiation uses positive ERA5 GHI values only. For precipitation, values below 0.1 mm define a separate dry regime, and P50, P75, and P95 are computed from wet reference hours only. Complete definitions and meteorological labels are given in Tables S2 and S3. This local-climatology construction follows established practice for climate extremes (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander, 2013).

Each verifying observation y is assigned to a regime by comparison with the thresholds for its station, calendar day, and UTC hour.

2.4 BIAS CORRECTION

Persistent model-wide biases can dominate forecast errors and obscure how performance changes across observed-value regimes. To reduce their influence, we apply a causal rolling bias correction before computing skill. For wind, temperature, and daytime solar radiation, the correction is additive. For model m , week w , and valid UTC hour h , let

$$\mathcal{D}_{m,w,h} = \left\{ (f_m(s, t_0, \tau), y(s, t_0 + \tau)) : \begin{array}{l} \text{the forecast–observation pair is available,} \\ t_0 + \tau \text{ lies in the four weeks preceding week } w, \\ \text{hour}_{\text{UTC}}(t_0 + \tau) = h \end{array} \right\}. \quad (2)$$

Thus, $\mathcal{D}_{m,w,h}$ pools matched forecast–observation pairs across European stations, initialisations, and leads at the same valid UTC hour. We estimate the bias as

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{m,w,h}|} \sum_{(f,y) \in \mathcal{D}_{m,w,h}} (f - y), \quad \tilde{f}_m(s, t_0, \tau) = f_m(s, t_0, \tau) - \bar{b}_{m,w,h(t_0+\tau)}. \quad (3)$$

For ensemble models, the same shift is applied to every member.

An additive correction is unsuitable for zero-bounded, right-skewed precipitation. We instead apply a multiplicative correction

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left(\frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t), \quad (4)$$

where $\mathcal{P}_w$ contain forecasts and observations from the preceding four weeks, pooled across initialisations and leads. This multiplicative correction adjusts systematic amount biases while preserving zero forecasts. Both procedures use only prior information and do not alter observation-defined regime membership or force the conditional error within each regime to zero.

2.5 EVALUATION PROTOCOL

We assess forecast accuracy relative to the high-resolution deterministic ECMWF IFS forecast. For a given variable, let $c = (s, t_0, \tau)$ denote an evaluation cell at station s , initialisation time t_0 , and lead time τ , verified at $t_0 + \tau$. For model m , let $\mathcal{C}_m$ denote the set of cells for which the bias-corrected forecast, the IFS forecast, and the verifying observation are all available. Each comparison therefore uses forecast–observation pairs matched in station, initialisation time, lead time, and valid time.

For the retained cells $\mathcal{C}_m$, MAE skill relative to IFS is

$$S_{m,\mathcal{C}_m} = 100 \left(1 - \frac{\text{MAE}_{m,\mathcal{C}_m}}{\text{MAE}_{\text{IFS},\mathcal{C}_m}} \right). \quad (5)$$

Positive skill indicates lower MAE than IFS, whereas negative skill indicates higher MAE. Initialisation times are restricted to the 00, 06, 12, and 18 UTC cycles.

The evaluation covers thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK, and it is organised into two tracks. Track A covers 1 September 2025 to 30 June 2026. Wind and temperature are evaluated at 6-hourly leads from 6 to 48 h, whereas solar radiation and hourly precipitation are evaluated at all leads from 1 to 48 h. Track B covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU on a common hourly schedule at leads from 1 to 48 h; EPT-2.1 Helios is additionally included for solar radiation. Italy and Poland are excluded from precipitation because of missing observations.

Forecasts are bilinearly interpolated to station locations (Molinaro et al., 2026). Ensemble models are evaluated through their means, so the analysis assesses point-forecast accuracy rather than probabilistic calibration or ensemble spread.

It should be observed that none of the models was trained or fine-tuned on the verification observations. Historical station errors enter only through the causal bias-correction procedure described above. Uncertainty in aggregated skill is quantified using a leave-one-month-out jackknife, with error bars representing one standard error.

3 RESULTS

3.1 SKILL ACROSS THE OBSERVED DISTRIBUTION

Track A results are reported in Figures 1 and 2. Figure 1 shows MAE skill relative to IFS by observed-value regime for wind, temperature, and solar radiation, while Figure 2 shows the corresponding precipitation results.

Tail skill does not follow a uniform AI–physical ordering. EPT-2.1 Europa leads the wind comparison overall and remains competitive with ICON Global in the gale-force regime. EPT-2 HRRR and EPT-2.1 Europa perform most strongly for temperature, including in the heat regime. By contrast, AIFS and AIFS ENS remain below IFS in the wind and temperature tails despite the ensemble’s improved performance near the centre of the distribution. The largest heat deficit belongs to the NWP model GFS.

Solar radiation follows a different pattern. EPT-2.1 Helios leads overall and under overcast and near-clear-sky conditions but falls below IFS under typical irradiance, where EPT-2 Reasoning performs most strongly. Its advantage is therefore concentrated in the tails rather than distributed uniformly across conditions.

For precipitation, the clearest gains occur in the middle of the wet-hour distribution (Figure 2). EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS from moderate through high rain, whereas the physical global models remain below the reference. These gains narrow at the heaviest intensities. Europa is the exception: its positive all-conditions score comes from dry hours, while it is worse than IFS in every wet-hour regime.

We additionally evaluate detection of heavy-rain events using the local wet-hour P95 threshold. For each forecast–observation pair, the observation and forecast are each labelled a heavy-rain event if their hourly accumulation exceeds the same local wet-hour P95 threshold. All evaluated precipitation models forecast fewer P95 exceedances than observed. Europa and ICON Global nevertheless achieve a slightly higher critical success index than IFS (Figure S7; Table S24).

We further assess the robustness of the Track A results to forecast lead, the inclusion of short-range forecasts, and country-level aggregation. The lead-time and 1–12 h analyses preserve the principal model ordering (Appendix A.2). Country-level results also show that the main temperature and wind patterns occur across multiple countries rather than being dominated by one country (Appendix A.5).

3.2 SHARED CONDITIONAL-ERROR PATTERN

Figure 3 reports conditional bias at 48 h by observed-value regime. Across all four variables, AI and NWP models, including IFS, show the same sign pattern: bias is positive for low observations and negative for high observations. The shift across regimes is larger than the differences between models at a fixed regime. This provides further evidence that conditional-error behaviour does not differ systematically between AI and NWP models, despite differences in relative MAE.

3.3 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

To complement the climatological-regime analysis, we evaluate five externally documented record-breaking temperature events and seven major European windstorms. Event windows and affected-country footprints were specified from external reports before scoring. A forecast sample is retained

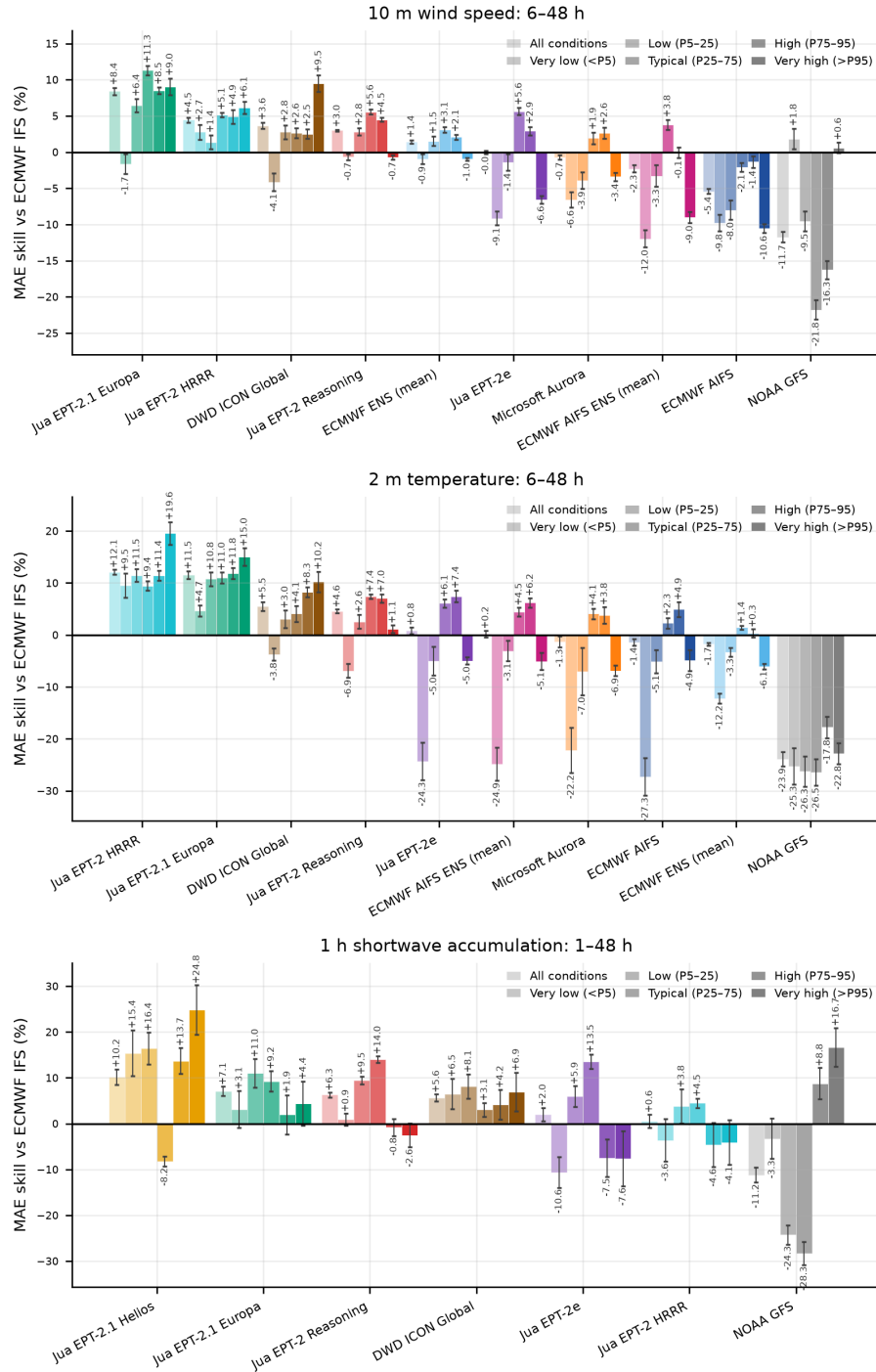


Figure 1: Track A MAE skill relative to ECMWF IFS by observed-value regime, with ± 1 jackknife SE. Wind and temperature use 6–48 h leads; solar radiation uses 1–48 h.

only when IFS and every Track A model included in the comparison share the same station, valid time, initialisation, and 6-hourly lead from 6 to 48 h. Each model therefore uses identical event windows and forecast–observation samples. Full event definitions, provenance, scoring details, and numerical results are reported in Appendix A.6 (Tables S25 and S26).

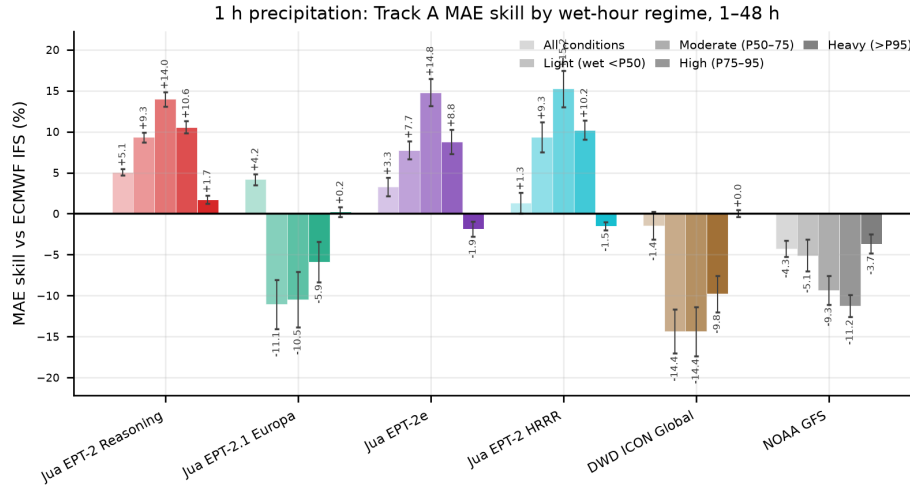


Figure 2: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime over 1–48 h, with ± 1 jackknife SE.

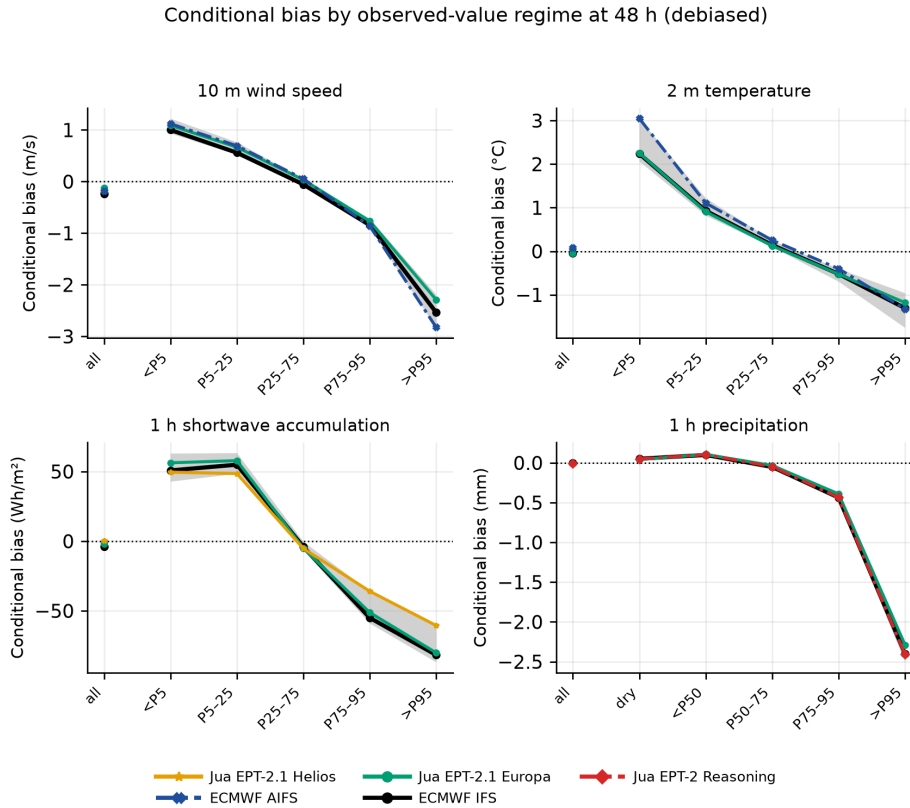


Figure 3: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models; selected models are highlighted.

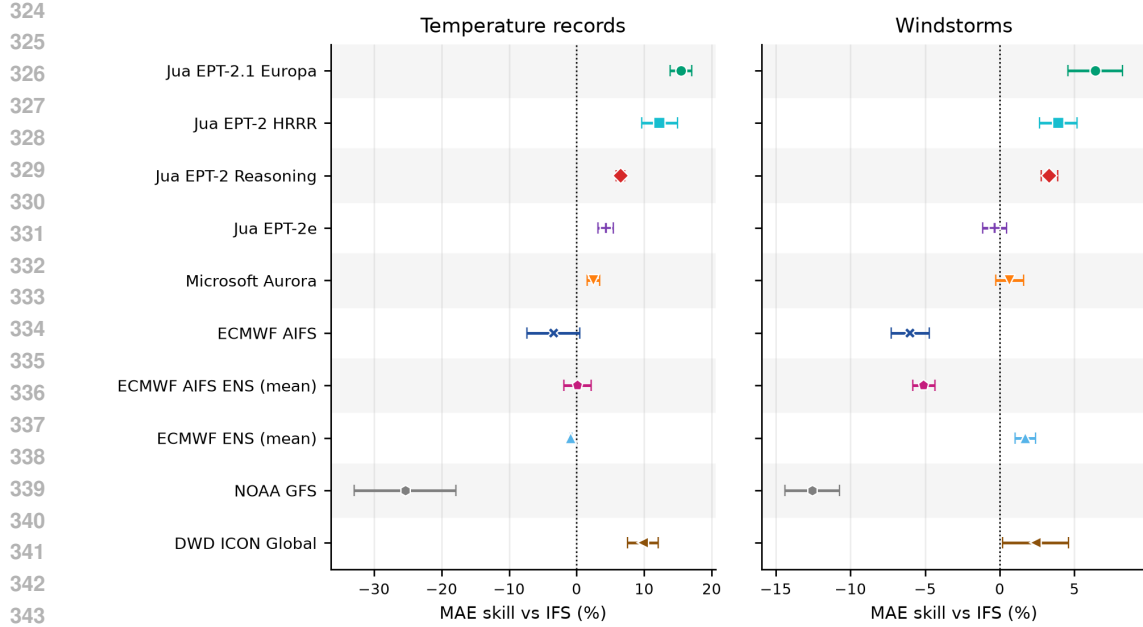


Figure 4: MAE skill relative to ECMWF IFS during five documented record-breaking temperature events (left; $n = 231,065$) and seven major European windstorms (right; $n = 123,847$), aggregated over common station/valid-time/lead samples from 6–48 h after bias correction. Error bars show one leave-one-event-out jackknife SE.

Figure 4 reports MAE skill relative to IFS during the documented windstorms and temperature events. Europa has the largest positive skill in both sets of events. During the temperature events, performance varies substantially across models, with GFS showing the largest deficit. During the windstorms, most models remain close to IFS, whereas AIFS, AIFS ENS, and GFS remain below the reference. These source-defined cases therefore support the model-specific pattern found in the climatological-regime analysis, rather than a uniform AI-specific loss of skill during extremes.

3.4 REGIONAL COMPARISON

Finally, we use Track B to compare EPT-2.1 Europa and EPT-2 HRRR directly with ICON-EU over their shared European domain from March to June 2026. Figure 5 reports MAE skill relative to IFS by observed-value regime. Europa retains the strongest overall wind estimate, while ICON-EU is competitive with the leading AI models in the wind and temperature tails. EPT-2.1 Helios has the clearest solar advantage.

The supplementary Track B panels resolve these comparisons by lead time and over the 1–12 h range; neither changes this interpretation (Appendix A.3). Complete numerical values are reported in Appendix A.4.

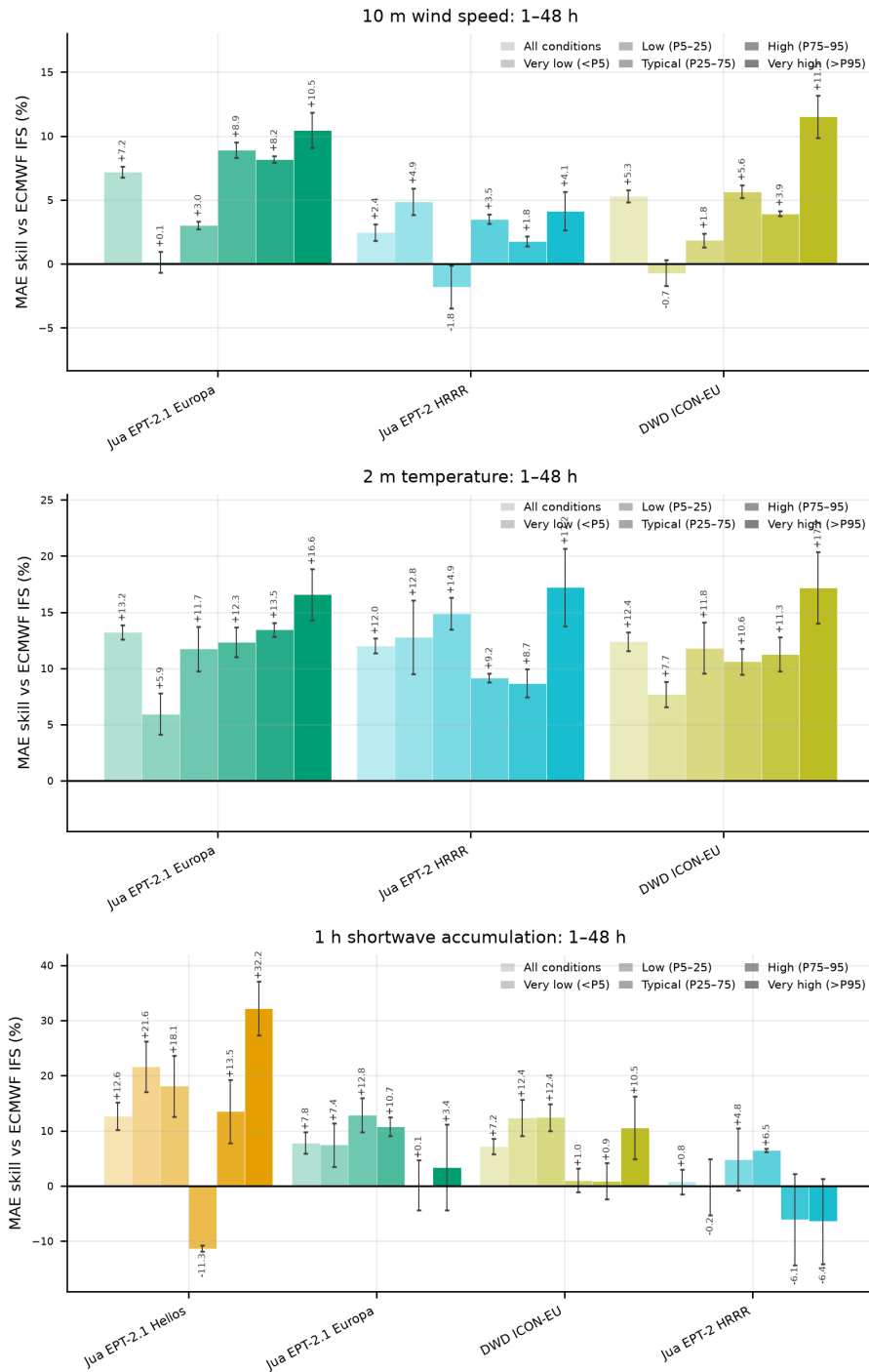


Figure 5: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

4 DISCUSSION

The results distinguish between two questions: whether a model underestimates an extreme observation, and whether it is less accurate than IFS when that observation occurs.

Across deterministic, probabilistic, and generative AI models, as well as NWP models, low observations are generally overpredicted and high observations are underpredicted. This shared conditional-error pattern does not yield a consistent class-level ordering. Instead, differences emerge at the level of individual systems: AIFS and AIFS ENS perform worse than IFS in the wind and temperature tails, whereas EPT-2 HRRR and EPT-2.1 Europa remain competitive in the gale-force and heat regimes. The NWP results likewise vary, with GFS showing the largest heat deficit while ICON-EU is competitive with the leading regional AI models in the matched Track B comparison. The source-defined temperature events and windstorms show the same model-specific pattern on common event samples: the leading regional systems remain ahead of IFS, whereas the two AIFS variants and GFS lose skill in at least one event set.

Relative accuracy also depends on the observed regime and variable. For example, EPT-2.1 Helios is more accurate than IFS in both solar tails but less accurate under typical irradiance. Similarly, EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS for moderate to high precipitation, although their advantage narrows for the heaviest events.

Within this scope, the results do not support a general weakness of AI weather forecasting at extremes.

4.1 LIMITATIONS

The comparisons identify differences in performance, but not their causes. The systems differ in architecture, training objective, resolution, training data, initialisation, cadence, and domain (Bonavita, 2024; Subich et al., 2025); consequently, their performance cannot be attributed independently to AI versus NWP, regression versus generative training, or global versus regional resolution. Moreover, evaluating ensembles through their means assesses point-forecast accuracy, rather than calibration, spread, or event probabilities.

ETHICS STATEMENT

The authors have a competing interest with respect to some of the evaluated forecast systems; details are withheld for anonymity and will be disclosed in the final version. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the configuration files, the derived aggregates behind every reported number, and the extraction, scoring, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates; station-level observations and raw forecast fields are not redistributed. The package is provided as anonymised supplementary material.

AI USE STATEMENT

Generative AI tools were used to review and refine the experimental methodology; to implement extraction, quality-control, scoring, aggregation, and figure code, including the cleaning and reformatting of observation and forecast data; to inspect and interpret generated results; to search, summarise, and verify related literature and format references; and to draft, structure, and edit manuscript text. They were not used to generate synthetic data, to alter individual observation or forecast values, or to prove mathematical claims. The authors reviewed and tested all AI-assisted code, verified the reported numbers against the aggregate data, and take full responsibility for the final content.

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A SUPPLEMENTARY MATERIAL

This supplement provides the model and regime definitions omitted from the main text, lead-time and short-range robustness checks for Track A, the matched regional comparison in Track B, complete numerical tables, and country-level regime profiles.

A.1 EVALUATION SETUP

Surface wind and temperature observations were obtained through Synoptic Data from selected operational observing networks, including SYNOP, WIS2, ASOS/AWOS, and METAR. Solar-radiation observations were compiled from ARPALomb, BSRN, DMI, DWD, FMI, Frost, GeoSphere Austria, KNMI, Météo-France, MeteoSwiss, RMI, and SMHI.

Table S1 identifies the forecasting systems and their broad model classes. Tables S2 and S3 give the observed-value intervals and meteorological labels used throughout the evaluation.

Table S1: Forecast models included in the evaluation. ECMWF IFS is the reference model.

Model	Model class	Forecast type	Domain
ECMWF IFS (reference)	Physical	Deterministic	Global
ECMWF ENS	Physical	Ensemble	Global
NOAA GFS	Physical	Deterministic	Global
DWD ICON Global	Physical	Deterministic	Global
DWD ICON-EU	Physical	Deterministic	Regional
ECMWF AIFS	Regression AI	Deterministic	Global
Microsoft Aurora	Regression AI	Deterministic	Global
Jua EPT-2 Reasoning	Regression AI	Deterministic	Global
Jua EPT-2e	Regression AI	Ensemble	Global
Jua EPT-2.1 Helios	Regression AI (solar)	Deterministic	Regional
ECMWF AIFS ENS	Probabilistic AI	Ensemble	Global
Jua EPT-2 HRRR	Generative AI	Ensemble	Regional
Jua EPT-2.1 Europa	Generative AI	Ensemble	Regional

Table S2: Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the percentiles refer to positive daytime GHI.

Regime	Definition	Wind	Temperature	Solar
All	–	all conditions	all conditions	all daytime
Very low	$y < P5$	calm	severe cold	heavily overcast
Low	$P5 \leq y < P25$	light air	cold	overcast
Typical	$P25 \leq y < P75$	moderate	moderate	mixed sky
High	$P75 \leq y < P95$	fresh to strong	warm	mostly sunny
Very high	$y \geq P95$	gale force	heat	near clear sky

Table S3: Observed-value regimes for hourly precipitation. Wet-hour percentiles are conditional on ERA5 accumulations of at least 0.1 mm.

Regime	Definition	Interpretation
All	–	all hours
Dry	$y < 0.1$ mm	dry hour
Light	$0.1 \text{ mm} \leq y < P50$	light rain
Moderate	$P50 \leq y < P75$	moderate rain
High	$P75 \leq y < P95$	heavy rain
Heavy	$y \geq P95$	most intense hours

A.2 TRACK A LEAD-TIME, SHORT-RANGE, AND PRECIPITATION ROBUSTNESS

Figures S1, S2, S3, and S4 show skill as a function of forecast lead. Figure S5 repeats the regime comparison over 1–12 h, while Table S4 reports heat-regime skill by country. Figure S6, Table S5, and Figure S7 provide the short-range, full-horizon, and categorical precipitation checks; short-range precipitation values are in Table S12.

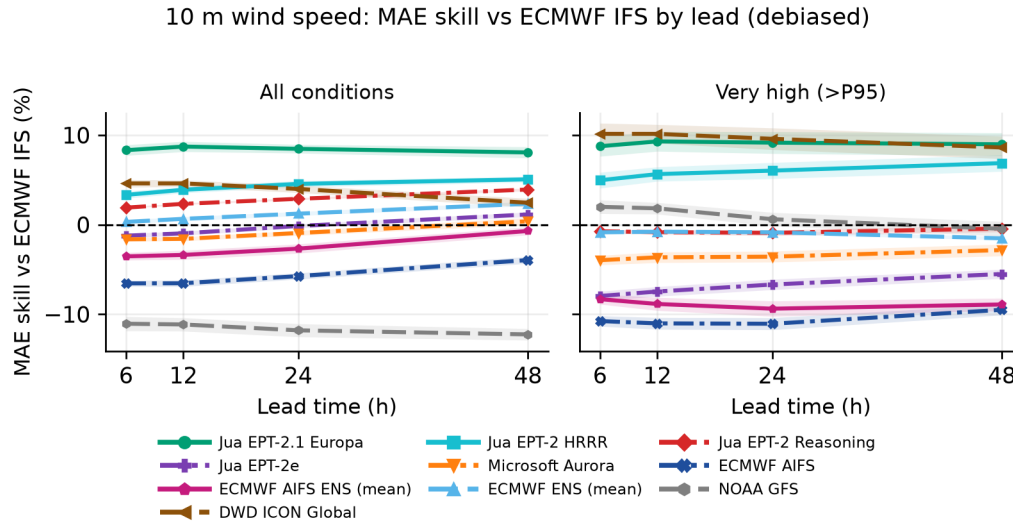


Figure S1: Track A MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

Table S4: Track A MAE skill (%) by country in the heat regime ($y \geq P95$), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF AIFS ENS (mean)	+5.3	-3.1	-6.7	-10.6	-12.6	-3.1	-3.9	+5.1	+4.6	-3.2	-38.7	+2.1	+4.4
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU [†]	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

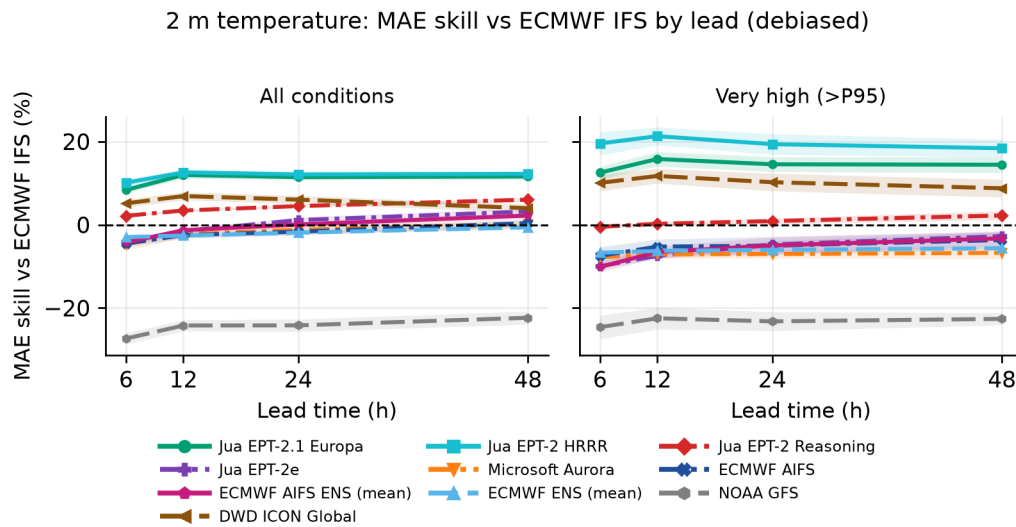


Figure S2: Track A MAE skill relative to ECMWF IFS by lead time for 2 m temperature. The left panel shows all conditions; the right panel shows the heat regime, $y \geq \text{P95}$. Shading denotes ± 1 jackknife SE.

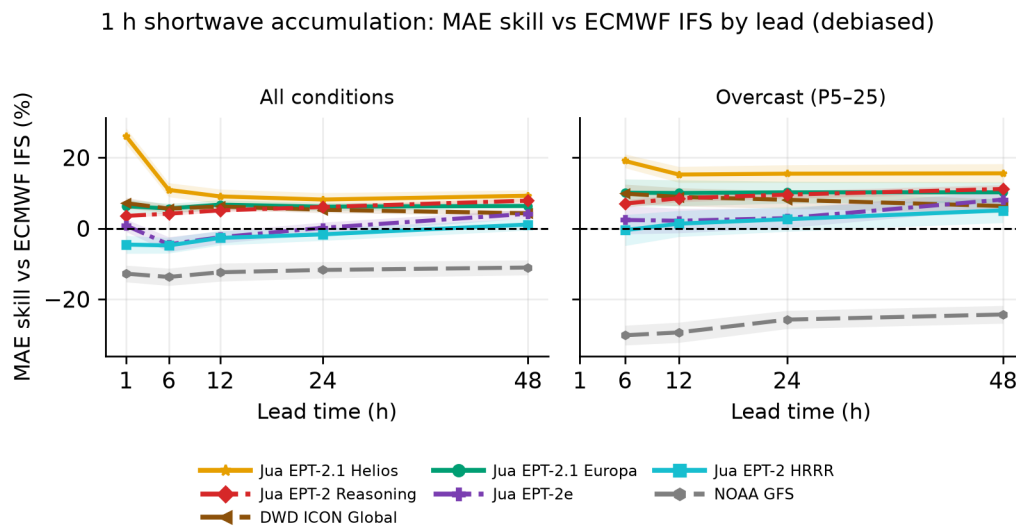


Figure S3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime, $P_5 \leq y < P_{25}$. AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes ± 1 jackknife SE.

Table S5: Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (± 0.4)	+2.2 (± 1.3)	+9.3 (± 0.6)	+14.0 (± 0.9)	+10.6 (± 0.8)	+1.7 (± 0.5)
Jua EPT-2 HRRR	+1.3 (± 1.3)	-6.1 (± 3.3)	+9.3 (± 1.8)	+15.2 (± 2.2)	+10.2 (± 1.2)	-1.5 (± 0.5)
Jua EPT-2e	+3.3 (± 1.1)	+1.5 (± 4.8)	+7.7 (± 1.1)	+14.8 (± 1.7)	+8.8 (± 1.5)	-1.9 (± 0.9)
Jua EPT-2.1 Europa	+4.2 (± 0.7)	+21.0 (± 2.7)	-11.1 (± 3.0)	-10.5 (± 3.4)	-5.9 (± 2.5)	+0.2 (± 0.6)
NOAA GFS	-4.3 (± 1.0)	-0.2 (± 1.8)	-5.1 (± 2.0)	-9.3 (± 1.8)	-11.2 (± 1.4)	-3.7 (± 1.2)
DWD ICON Global	-1.4 (± 1.7)	+8.3 (± 2.7)	-14.4 (± 2.7)	-14.4 (± 3.0)	-9.8 (± 2.2)	+0.0 (± 0.4)

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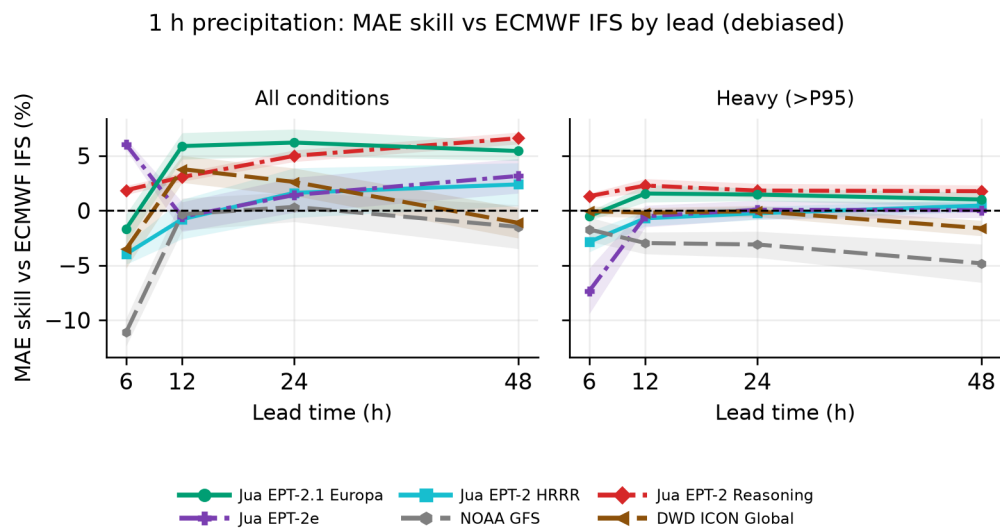


Figure S4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

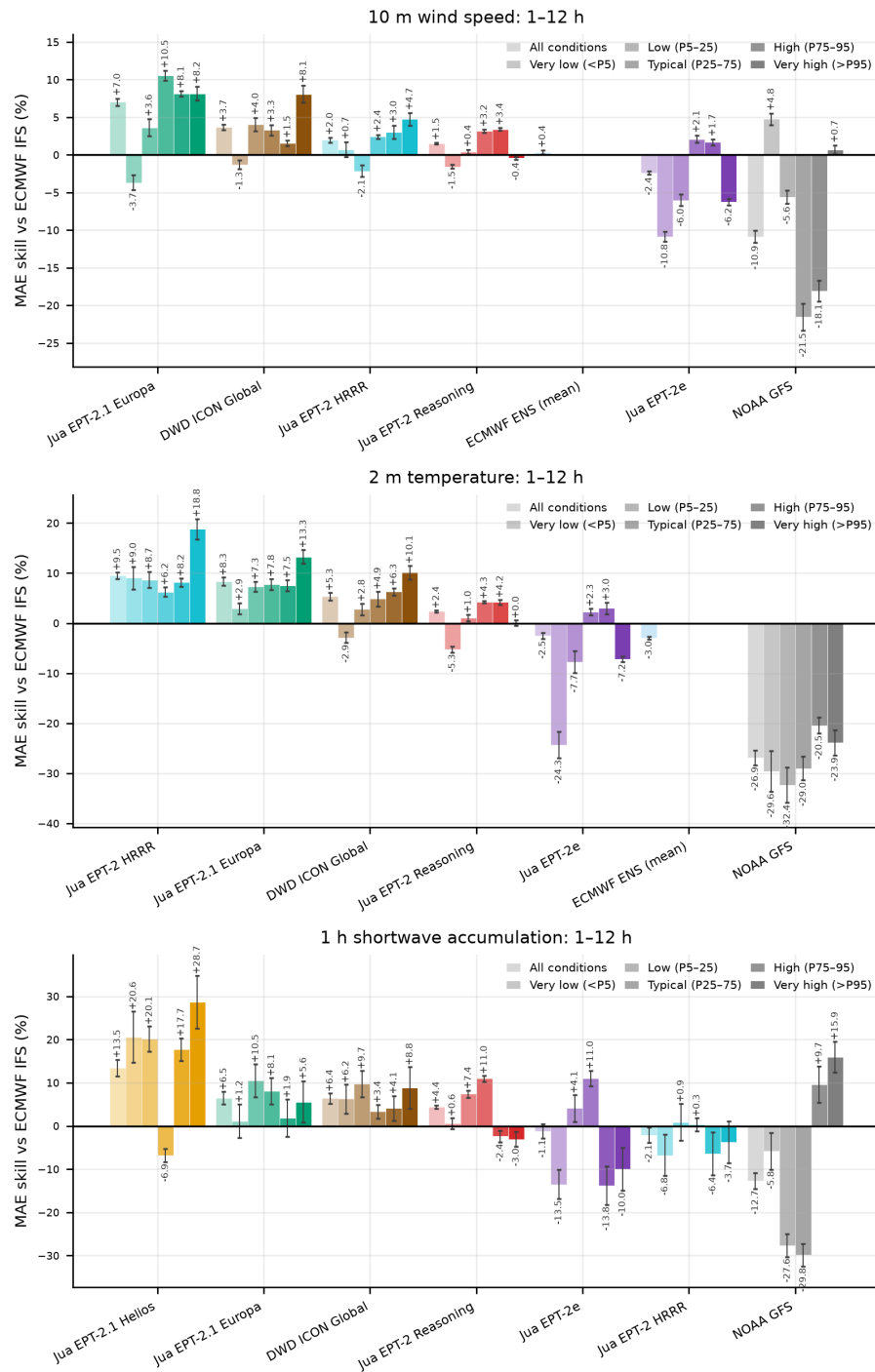


Figure S5: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with ± 1 jack-knife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

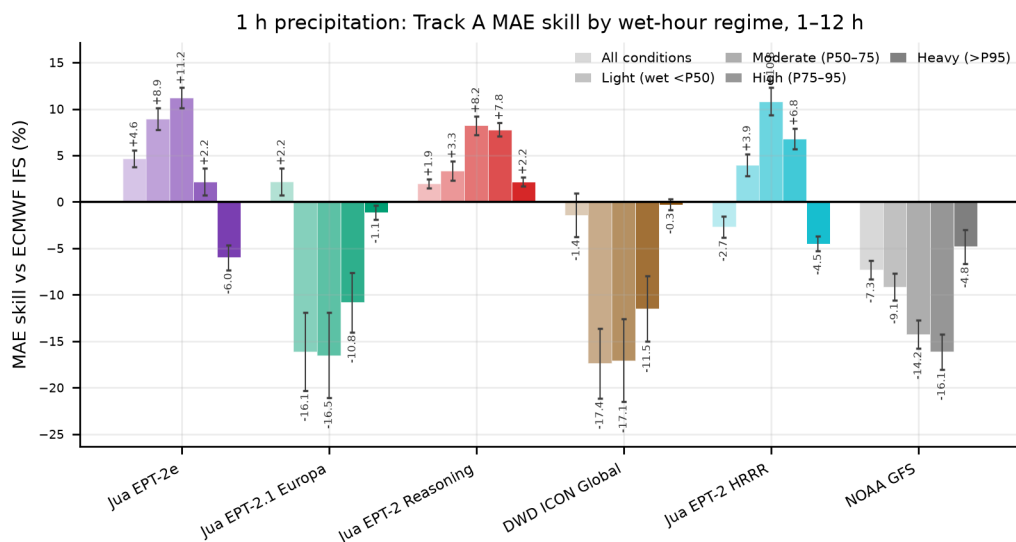


Figure S6: Track A MAE skill relative to ECMWF IFS for precipitation by observed-value regime over 1–12 h, with ± 1 jackknife SE

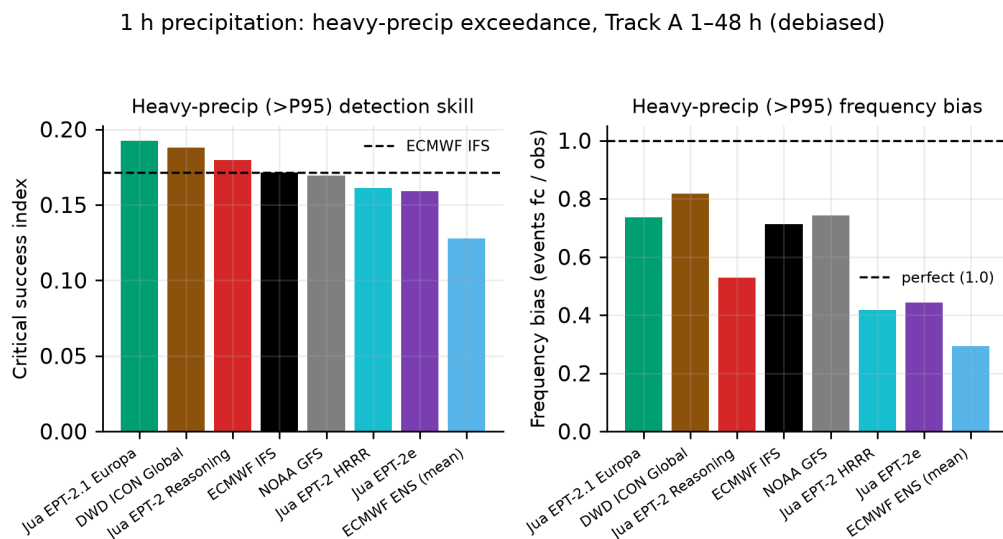


Figure S7: Detection of heavy precipitation ($y \geq P95$) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

A.3 MATCHED REGIONAL COMPARISON

Track B robustness beyond the full-horizon comparison in the main text is shown in Figure S10 for 1–12 h and in Figure S8 as a function of lead. Figures S11 and S9 give the corresponding precipitation results.

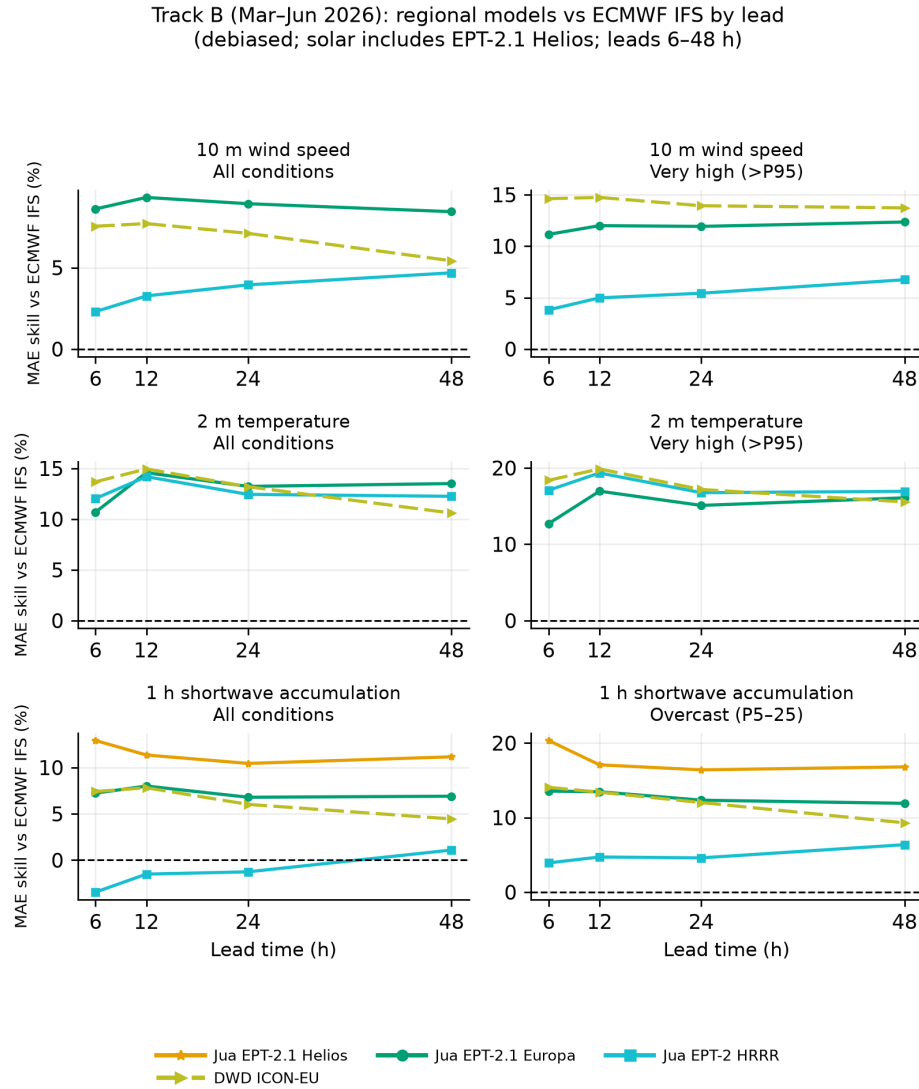


Figure S8: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes ± 1 jackknife SE.

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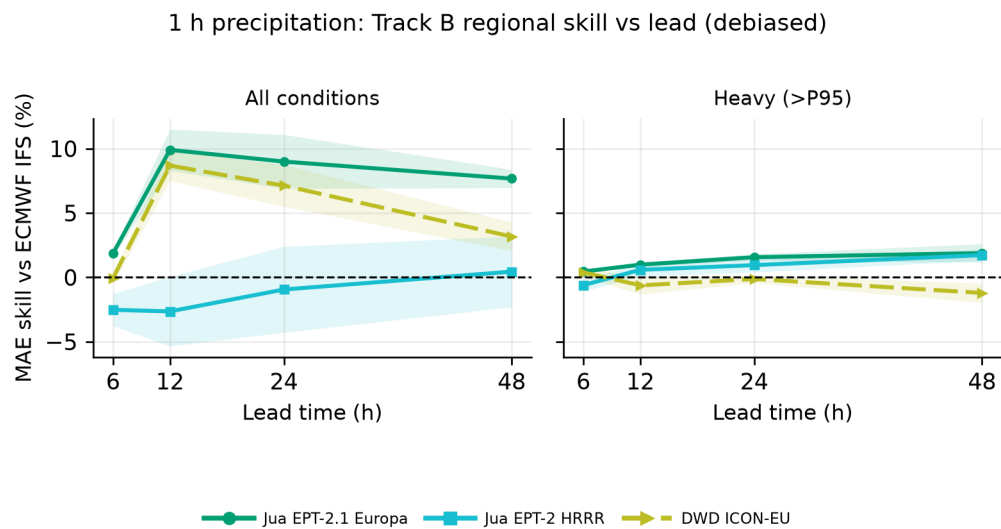


Figure S9: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

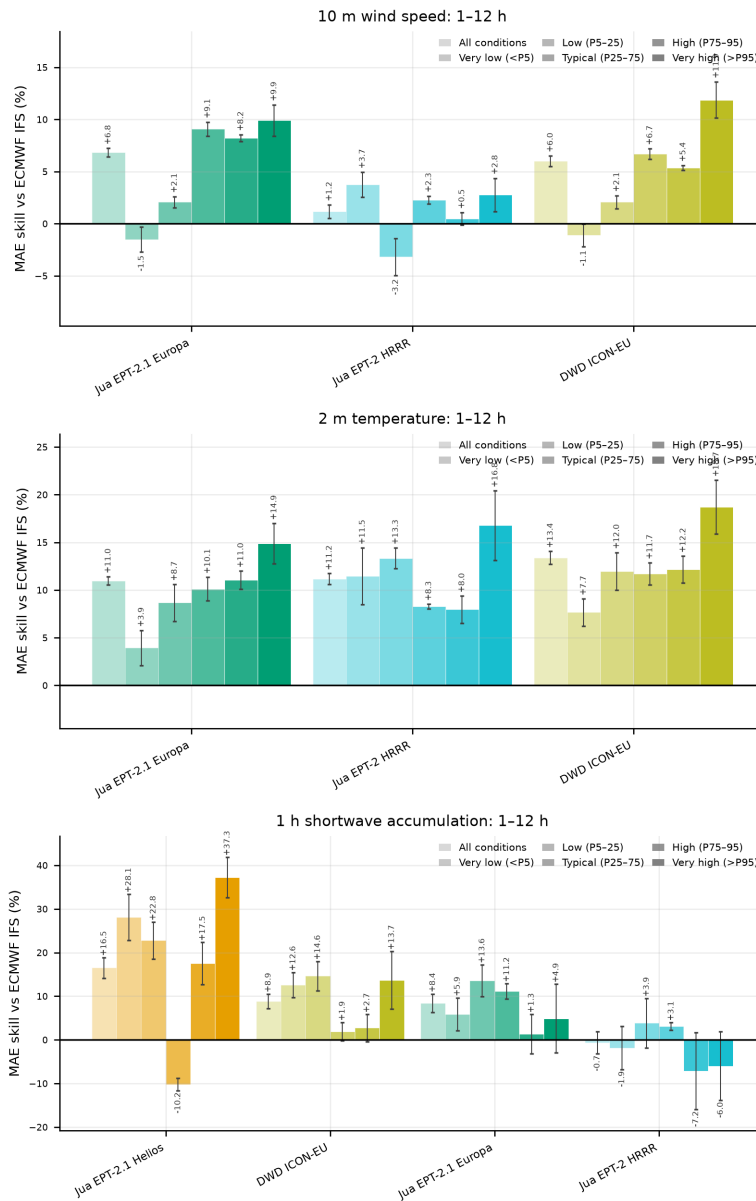


Figure S10: Track B MAE skill relative to ECMWF IFS for wind, temperature and solar by observed-value regime over 1–12 h, with ± 1 jackknife SE

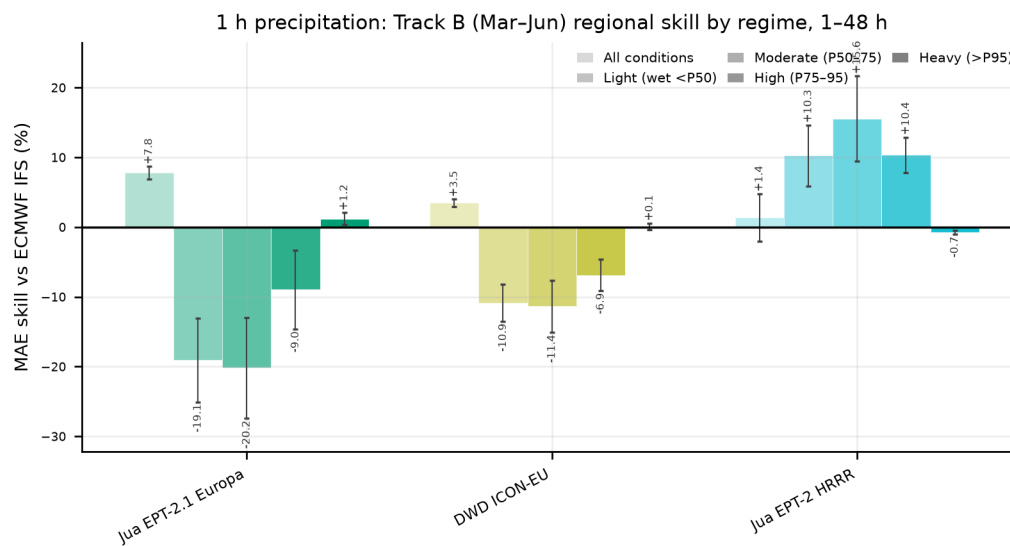


Figure S11: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE.

A.4 COMPLETE NUMERICAL RESULTS

Track A full-horizon regime values are in Tables S6, S7, and S8; 1–12 h values are in Tables S9, S10, S11, and S12; per-lead values are in Tables S13, S14, and S15. Track B regime values are in Tables S16, S17, S18, and S23; its per-lead values are in Tables S19, S20, S21, and S22. Table S24 gives the heavy-precipitation detection metrics.

Table S6: Track A MAE skill (%), 6–48 h, by regime – Figure 1, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF AIFS ENS (mean)	-2.3 (0.5)	-12.0 (1.2)	-3.3 (1.5)	+3.8 (0.7)	-0.1 (0.7)	-9.0 (0.7)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S7: Track A MAE skill (%), 6–48 h – Figure 1, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF AIFS ENS (mean)	+0.2 (0.6)	-24.9 (3.2)	-3.1 (2.0)	+4.5 (0.9)	+6.2 (0.9)	-5.1 (1.7)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S8: Track A MAE skill (%), 1–48 h hourly – Figure 1, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S9: Track A MAE skill (%), 1–12 h – Figure S5, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S10: Track A MAE skill (%), 1–12 h – Figure S5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S11: Track A MAE skill (%), 1–12 h – Figure S5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S12: Track A precipitation MAE skill (%), 1–12 h (Figure S6). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S13: Track A MAE skill (%) by lead – Figure S1, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
AIFS ENS	-3.5	-8.3	-3.4	-8.9	-2.7	-9.4	-0.7	-8.9
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S14: Track A MAE skill (%) by lead – Figure S2, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
AIFS ENS	-4.3	-10.1	-1.3	-6.6	+0.1	-5.0	+2.2	-3.3
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S15: Track A MAE skill (%) by lead – Figure S3, solar (Ovc. = overcast, P5–P25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S16: Track B wind MAE skill (%), 1–48 h, by regime (Figure 5).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S17: Track B MAE skill (%), 1–48 h – Figure 5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S18: Track B MAE skill (%), 1–48 h – Figure 5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S19: Track B MAE skill (%) by lead – Figure S8, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S20: Track B MAE skill (%) by lead – Figure S8, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S21: Track B MAE skill (%) by lead – Figure S8, solar (Ovc. = overcast, P5–P25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S22: Track B precipitation MAE skill (%) by lead (Figure S9). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

Table S23: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure S11. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (± 0.9)	+29.3 (± 1.4)	-19.1 (± 6.0)	-20.2 (± 7.2)	-9.0 (± 5.7)	+1.2 (± 0.8)
Jua EPT-2 HRRR	+1.4 (± 3.4)	-3.6 (± 9.7)	+10.3 (± 4.4)	+15.6 (± 6.1)	+10.4 (± 2.5)	-0.7 (± 0.3)
DWD ICON-EU	+3.5 (± 0.5)	+15.3 (± 0.3)	-10.9 (± 2.7)	-11.4 (± 3.8)	-6.9 (± 2.2)	+0.1 (± 0.5)

Table S24: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure S7. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

A.5 COUNTRY-LEVEL REGIME PROFILES

Figures S12, S13, S14, and S15 show how the March–June regime-skill patterns vary across countries for the mixed regional/global comparison set. Countries without sufficient matched observations for a variable are omitted.

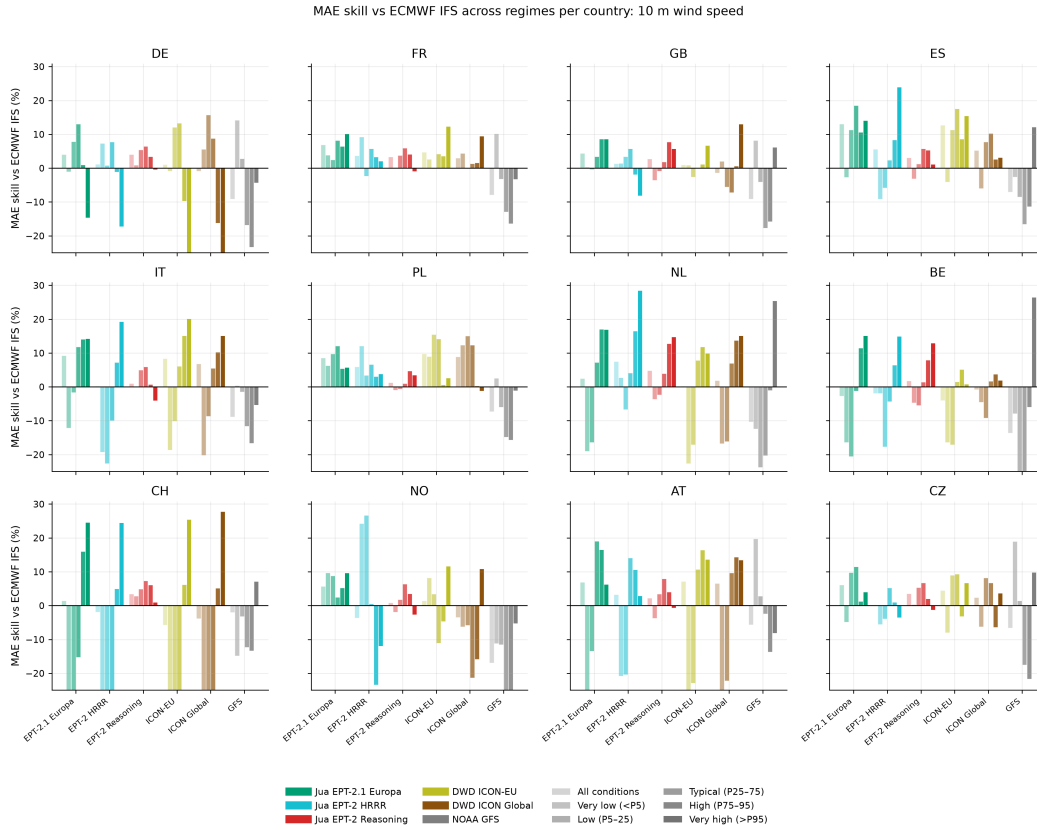


Figure S12: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set.

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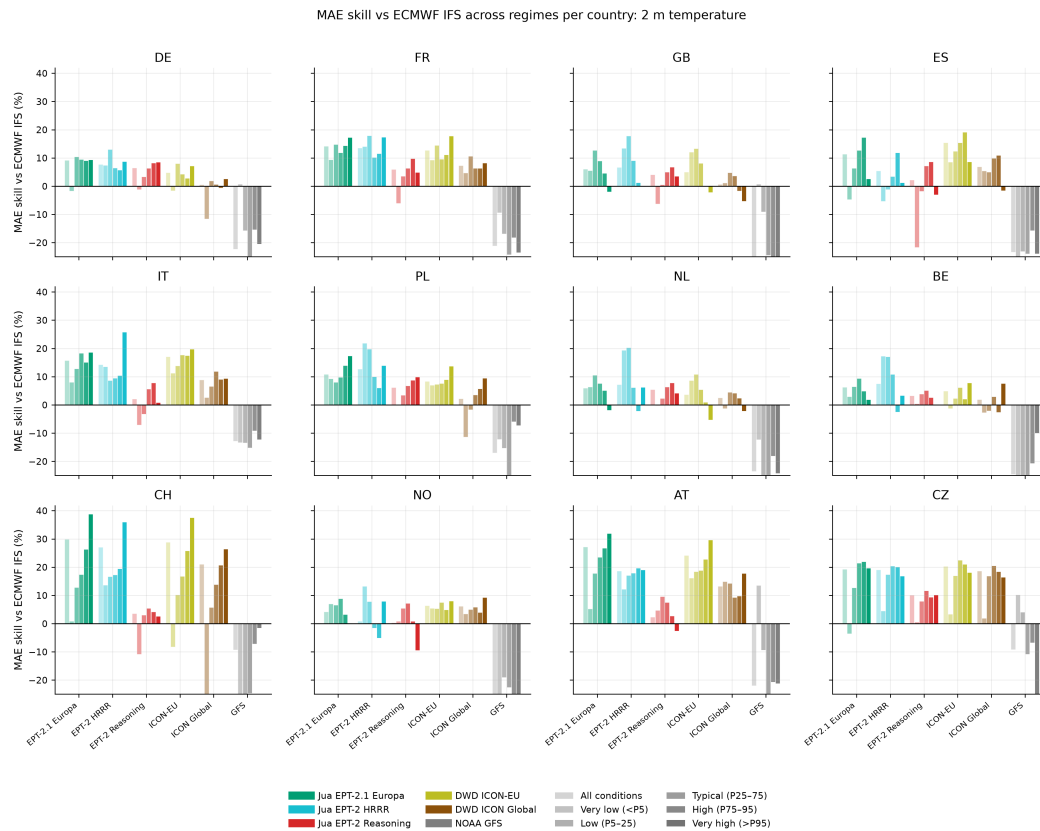


Figure S13: As Figure S12, for 2 m temperature.

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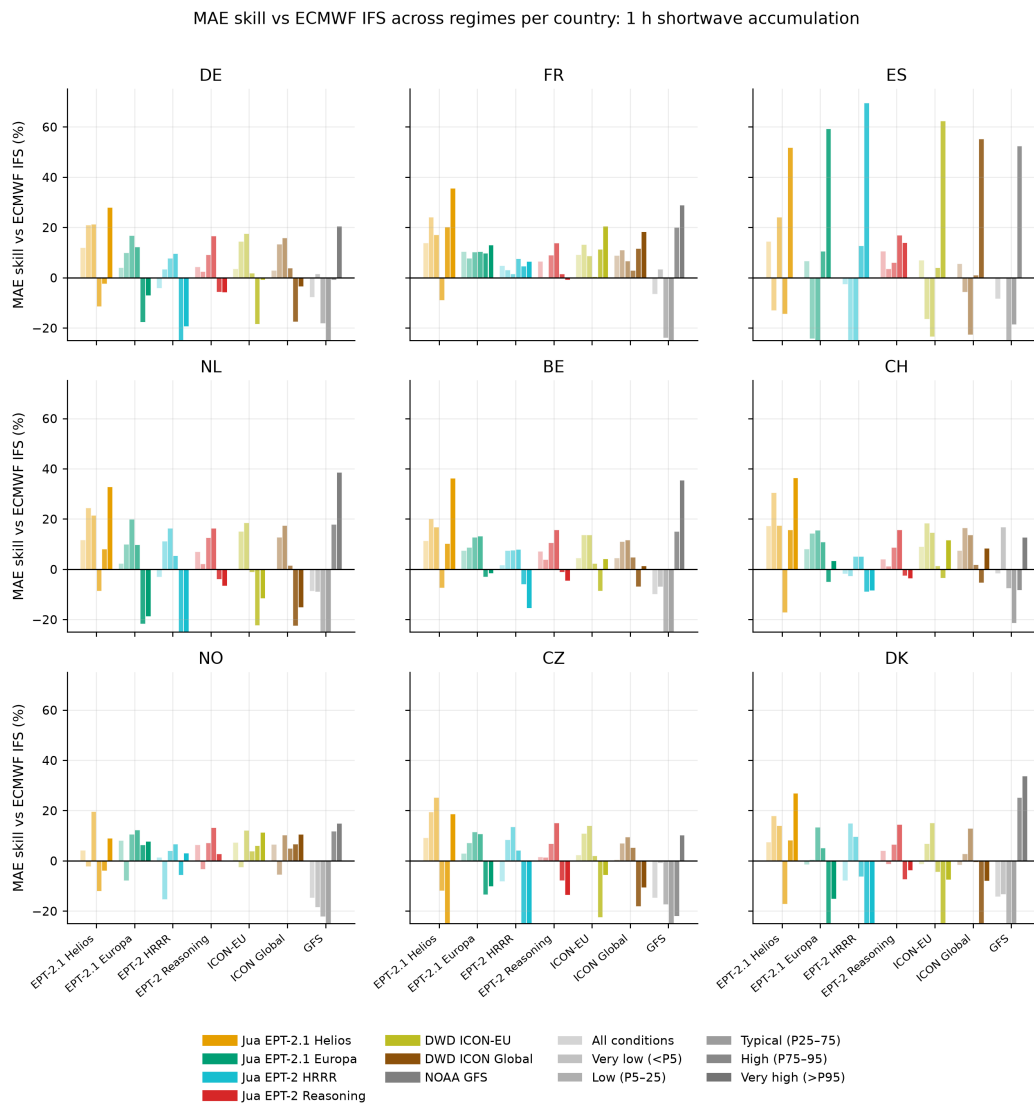


Figure S14: As Figure S12, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 2.2).

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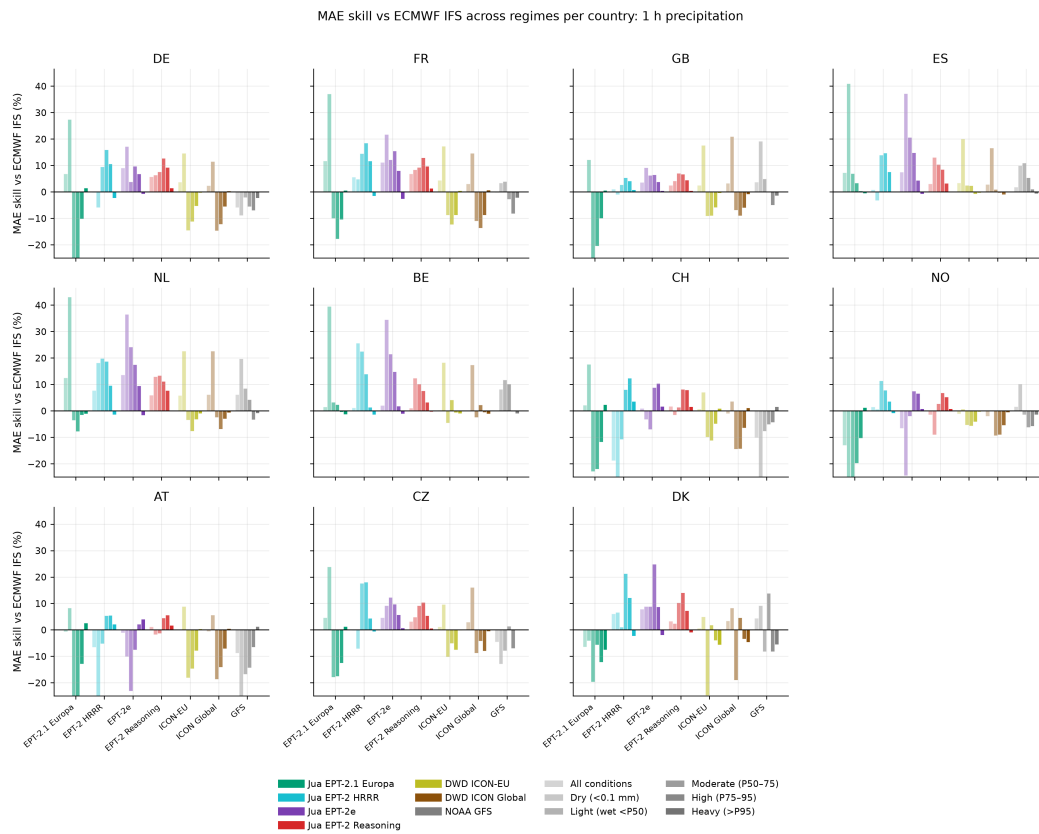


Figure S15: Per-country MAE skill by precipitation regime for the mixed regional/global model set. IT and PL have no gauges in the precipitation panel.

A.6 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

Table S25 lists the externally documented temperature-record events and European windstorms used in the case-study analysis. Event windows and affected countries were fixed from the source catalogue before scoring. Every benchmark station in the source-reported countries is included at 00, 06, 12, and 18 UTC throughout the event window. The analysis is therefore conditioned on documented events, rather than on whether an individual station observation breaks a record.

For event e and lead $\ell \in \{6, 12, \dots, 48\}$ h, let $\mathcal{C}_{e,\ell}$ denote the station–valid-time pairs (s, t) in the event footprint for which ECMWF IFS and every Track A model included in the comparison provide a forecast initialised at $t - \ell$. Thus, each model is evaluated against the same observations at each lead. For ensemble models, f_m denotes the ensemble mean and $\tilde{f}_m$ the forecast after the causal bias correction of Section 2.4. We compute

$$\begin{aligned}\varepsilon_m(s, t, \ell) &= \tilde{f}_m(s, t | t - \ell) - y(s, t), \\ N &= \sum_e \sum_\ell |\mathcal{C}_{e,\ell}|, \\ \text{MAE}_m^{\text{event}} &= \frac{1}{N} \sum_e \sum_\ell \sum_{(s,t) \in \mathcal{C}_{e,\ell}} |\varepsilon_m(s, t, \ell)|, \\ S_m^{\text{event}} &= 100 \left(1 - \frac{\text{MAE}_m^{\text{event}}}{\text{MAE}_{\text{IFS}}^{\text{event}}} \right).\end{aligned}\tag{6}$$

The same observation may contribute at multiple leads because these are distinct forecasts from different initialisation times. For uncertainty, we delete one complete event at a time and report the jackknife standard error

$$\sqrt{\frac{E-1}{E} \sum_e \left(S_{m,(-e)}^{\text{event}} - \bar{S}_{m,(-\cdot)}^{\text{event}} \right)^2},$$

where $E = 5$ for temperature and $E = 7$ for wind. Complete pooled values and event-level uncertainty are reported in Table S26.

Table S25: Source-defined event catalogue. The accompanying reproduction package records the source URLs and selection basis for every event.

Variable	Event	Dates	Affected countries
Temperature	Central European December record warmth	2025-12-07–2025-12-09	NL, CZ
Temperature	Late-February European record warmth	2026-02-25–2026-02-27	FR, DE, ES, CZ
Temperature	Early-April record warm spell	2026-04-05–2026-04-08	FR, GB
Temperature	Western European May heatwave	2026-05-21–2026-05-30	FR, GB
Temperature	Late-June European record heatwave	2026-06-21–2026-06-28	DE, FR, GB, ES, PL, NL, CH, AT, CZ, DK
Wind	North Sea autumn storm	2025-09-15–2025-09-16	NL
Wind	Storm Amy	2025-10-03–2025-10-05	GB, FR, BE, NL, NO, DK, PL
Wind	Storm Benjamin	2025-10-23–2025-10-24	FR, NL, CH, GB
Wind	Storm Goretti	2026-01-08–2026-01-09	GB, FR, DE, NL
Wind	Storm Nils	2026-02-11–2026-02-13	FR, ES
Wind	Storm Marlis	2026-03-25–2026-03-26	CH, DE
Wind	Storm Deborah	2026-03-26–2026-03-27	FR, IT

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Table S26: Event-case MAE skill (%) relative to ECMWF IFS. Parentheses give leave-one-event-out jackknife SE. Temperature uses five events and 231,065 common samples; wind uses seven events and 123,847 common samples.

Model	Temperature events	Windstorms
Jua EPT-2.1 Europa	+15.4 (1.6)	+6.4 (1.8)
Jua EPT-2 HRRR	+12.3 (2.7)	+3.9 (1.3)
Jua EPT-2 Reasoning	+6.5 (0.6)	+3.3 (0.5)
Jua EPT-2e	+4.3 (1.1)	-0.3 (0.8)
Microsoft Aurora	+2.5 (0.9)	+0.7 (0.9)
ECMWF AIFS	-3.4 (3.9)	-6.0 (1.3)
ECMWF AIFS ENS (mean)	+0.1 (2.0)	-5.1 (0.7)
ECMWF ENS (mean)	-0.9 (0.1)	+1.7 (0.7)
NOAA GFS	-25.4 (7.5)	-12.6 (1.8)
DWD ICON Global	+9.8 (2.3)	+2.4 (2.2)